AUTUMN MID SEMESTER EXAMINATION-2025

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject Name: Engineering Economics
[Subject Code: HS30101]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 X 5]

   a) Explain the impact on the equilibrium price when supply decreases and demand remains unchanged. Illustrate your answer with an appropriate diagram.

   Ans: Equilibrium price increases.

   b) Write the conditions of consumer equilibrium.

   Ans: $P_x/P_y = MU_x/MU_y = \Delta Y/\Delta X$ (for two products X and Y)

   c) Write an expression for Marginal Revenue (MR) in terms of price elasticity of demand.

   Ans: $MR = P\left(1-\frac{1}{e}\right)$

   d) The price elasticity between product X and Y is +0.8. If the product Y got a 10% price rise with initial demand for product X at 200 units, find the new demand.

   Ans: 216 units

   e) G. Ranjan deposited $10000 in his savings account now at the rate of interest 8% compounding annually. At the end of fourth year he deposited another amount of $10000 in the same account as the rate of interest increased to 8.5% compounded annually. Find the maturity amount that he will receive at the end of 10th year.

   Ans: $38510.61

2. The demand and supply functions of a product are given as follows: $Q_d = 25{,}000 - 400P$ and $Q_s = 7000 + 1200P$.

   (i) Find the equilibrium price and quantity.

   (ii) After a certain time, consumers revised their demand function for this product as $Q_d = 22{,}000 - 400P$ due to emergence of new products in the market. Find out new equilibrium price and quantity after the change in demand. [5 Marks]

   Ans: Initial equilibrium: Price = 11.25, Quantity = 20500

   New equilibrium: Price = 9.375, Quantity = 18250

3. Define income elasticity of demand and explain how income elasticity differs for inferior goods, normal goods, necessary goods, and luxury goods. Suppose the income of a household increases by 20%. The demand for four products changes as follows:

- Rice: increases by 5%
- Air-conditioners: increases by 40%
- Bus travel: decreases by 10%
- Clothing: increases by 20%

Calculate the income elasticity of demand for each product and classify them. [5 Marks]

Ans: Income elasticity of demand (IE) measures how quantity demanded responds to a change in income.

- Rice: IE = 0.25 → Normal / Necessity
- Air-conditioners: IE = 2.00 → Luxury
- Bus travel: IE = −0.50 → Inferior
- Clothing: IE = 1.00 → Normal

4. An electric vehicle company records the sales over the last 6 years as follows:

Year	2019	2020	2021	2022	2023	2024
Sales	120	140	150	170	180	200

Fit the trend equation and estimate the sales of the company for the year 2026. [5 Marks]

Ans: $Y=160+39.47X$; $Y_{(2026)} = 357.35$

5. Ghanashyam Singh made an initial deposit now in his account to receive a future sum of $500000 at the end of 8 years. His account is growing at 8% interest annual compounding. His wife Kasturi Singh also wants $500000 at the end of same time period, but she can deposit an equivalent amount at the end of each year. She will get the same rate of interest in her account. Determine:

i) The initial deposit made by Ghanashyam

ii) The equivalent amount to be deposited by Kasturi Singh. [5 Marks]

Ans: i) Ghanashyam's single deposit now = $270134.44
ii) Kasturi's annual deposit = $47007.38

*** Best of Luck ***